

Partitioning a 100" Domed Cylinder in Abaqus/CAE

Part: 3D deformable **solid**. ~100" dia. cylinder wall (~3.5" thick, 30" tall) with an integral dome that meets the bottom edge of the wall and rises to the top edge. The dome face carries multiple 7"-dia. holes arranged on **staggered circular (bolt-circle) patterns at several radii**.

Goal:

1. Cut the part in half on the **XY plane** for symmetry.
2. Perform a **"shell revolve" partition** — revolve a **meridian section sketch** about the principal axis (360° on the full part, or 180° on the half) so a line traced through the **hole center-points** becomes a continuous ring/mid-surface partition.
3. Add **radial "sector" partitions** running from the **dome center out to each hole center** (one radial line per staggered hole).

Key Abaqus fact: there is **no dedicated "Revolve" partition button**. A revolved partition is created by sketching the section on a datum plane that contains the axis, then **revolving those sketch edges about a datum axis** with **Partition Cell** → **Extrude/Sweep Edges** (sweeping along a *circular edge* = the revolve). This guide uses that tool for the "shell revolve."

0. Conventions and assumptions

- **Revolution axis = vertical centerline**, assumed to be the global **Y-axis** (height direction). If your axis is Z, swap $Y \leftrightarrow Z$ throughout.
- Because the axis is Y, the **XY** and **YZ** principal planes both contain the axis. Your symmetry cut is on **XY**.

- Holes are drilled through the dome wall with their **centers on the dome mid-thickness surface**, so a single meridian line traced through the mid-thickness passes through **all** hole centers regardless of radius. Revolving that line is the "shell revolve."
- Work in the **Part** module, **Tools** → **Partition**, before meshing. Turn on **View** → **Part Display Options** → **Datum** so datums stay visible.

Order of operations (recommended): do the **360° shell revolve first on the intact part**, then the radial sectors, then the XY half-cut *last*. Revolving on the full part lets you sweep the **full circular edge** (most robust). The half-model (180°) variant is covered too — jump to §3B note.

1. Create the datums

1a. Datum axis (revolution centerline)

1. **Tools** → **Datum** → **Type: Axis** → **Method: Principal axis** → **Y-Axis** → **Apply**.
2. *(If the centerline is not a principal axis: Method **2 points**, pick the dome apex vertex and the base-center vertex.)*

1b. Datum plane through the axis (meridian sketch plane + symmetry plane)

1. **Tools** → **Datum** → **Type: Plane** → **Method: Principal plane** → **XY-Plane, Offset 0** → **OK**.

This one XY plane serves as both the meridian sketch plane (§3) and the symmetry cut plane (§4).

2. (Reference) the two partition tools you'll use

Need	Tool	Menu
Slice with a flat plane	Use datum plane	Tools → Partition → Cell → Use datum plane
Sketch edges on a flat face	Sketch (planar)	Tools → Partition → Cell → Sketch
Revolve those edges about the axis	Extrude/Sweep Edges	Tools → Partition → Cell → Extrude/Sweep edges
Radial cut on the curved face	Shortest path between 2 points	Tools → Partition → Face → Use shortest path

3. The "shell revolve" partition (meridian section sketch → revolve)

3A. Draw the meridian section sketch

Menu: Tools → Partition → **Cell** → **Sketch** (planar).

1. **Tools** → **Partition** → **Type = Cell** → **Method = Sketch**.
2. **Select the solid cell** to partition → **Done**.
3. When asked for the sketch plane, **pick the XY datum plane** (the meridian plane through the axis). Pick the datum axis (or a vertical edge) as the "vertical/up" reference so the section is oriented naturally.
4. You are now looking at the **meridian cross-section** of the dome + wall. In the Sketcher:
 - **Trace the dome outline:** use **Add** → **Sketch** → **Project**

Edges (or Project Reference) to bring the **inner and outer dome surface profiles** into the sketch as reference geometry. This gives you the true dome contour to work against.

- **Draw the "hole-center" line:** draw a spline/arc/polyline that runs **through the mid-thickness of the dome wall**, i.e. the contour that passes through the **center-points of every hole** (all radii). Because the holes are centered in the wall, this mid-contour hits every staggered hole center when revolved.
 - Constrain it to the dome contour: dimension its offset to **½ of the 3.5" wall (≈1.75")** from the inner (or outer) surface, or place its endpoints at the known hole-center radii **R₁, R₂, R₃...** (R = each bolt-circle diameter ÷ 2) and heights.
- *(If instead you want discrete rings only at each hole radius, draw a short radial tick crossing the wall at each R_i rather than one continuous mid-line.)*

5. Done → Create Partition.

You now have **edges lying on the meridian (XY) face** that trace the hole-center contour. These are the "section" you revolve.

3B. Revolve the section edges about the axis

Menu: Tools → Partition → **Cell** → **Extrude/Sweep edges** — this is the revolve.

1. **Tools** → **Partition** → **Type = Cell** → **Method = Extrude/Sweep edges**.
2. **Select the meridian edges** you just made (the hole-center contour) → **Done**.
3. Choose the **"Sweep along an edge"** option and **select a circular edge that encircles the axis** as the sweep path:
 - **Full 360° part:** pick the dome's **full outer circular edge** → the section revolves a complete 360°, producing a continuous ring/mid-surface partition through every hole center.
 - **Half model (180°):** if you've already cut on XY, pick the

semicircular (180°) edge instead → the section revolves 180° across the existing material.

4. Create Partition.

Do NOT pick the datum axis here. Selecting the axis as a *direction* makes Extrude/Sweep **translate** (straight extrude), not revolve. The revolve comes from choosing a **circular edge as the path**.

If Extrude/Sweep refuses to revolve cleanly (profile/path not compatible), use the robust fallback in §3C.

3C. Robust fallback for the rings — horizontal datum planes

Because the dome is a surface of revolution, **each hole-ring radius corresponds to a unique height** on the dome. A **horizontal plane** (normal to the axis) at that height cuts the dome in exactly the circle through those hole centers:

1. **Tools** → **Datum** → **Plane** → **Offset from principal plane** → **XZ-Plane**, enter the **height H_i** of each hole ring → OK (repeat per ring).
2. **Tools** → **Partition** → **Cell** → **Use datum plane**, select the cell, select each horizontal datum plane.

This gives clean, guaranteed circular ring partitions through the hole centers without depending on sweep behavior.

4. Radial "sector" partitions — dome center → each hole center

You asked for **one radial line through each hole center**. Since the pattern is **staggered**, each ring's holes sit at different azimuth angles, so each hole gets its own radial cut.

Preferred (fast, on the face): Tools → Partition → **Face** → **Use**

shortest path between 2 points.

1. **Tools → Partition → Type = Face → Method = Use shortest path between 2 points.**
2. **Pick the dome apex/center vertex**, then **pick a hole's center vertex** (or the two rim vertices bracketing it) → Abaqus lays a geodesic radial partition line across the curved dome face from center to that hole.
3. Repeat for each hole. The result is a set of radial "spokes" from the dome center to every hole center — the sectors you described.

Through-thickness sectors (if you need the cut through the full wall, not just the face): use rotated datum planes:

1. **Tools → Datum → Plane → Rotate from a plane about an axis:** pick the XY plane, pick the datum axis, enter the **azimuth angle of each hole** (measured from XY). Create one plane per hole angle.
2. **Tools → Partition → Cell → Use datum plane** with each rotated plane.
 - *Tip:* if the staggered angles are regular within a ring (e.g. holes every $360/N$), you only need N planes per ring; reuse them across rings where angles coincide.

Combined with the §3 rings, this builds a **polar grid centered on the dome**, with a node/partition landing on every hole center — ideal for seeding a clean mesh.

5. Cut the part in half on XY (do this LAST)

Menu: Tools → Partition → Cell → Use datum plane.

1. **Tools → Partition → Type = Cell → Method = Use datum plane.**
2. **Select the cell(s) → Done.**
3. **Select the XY datum plane → Create Partition.**

Partitioning only *subdivides*; it doesn't delete material. For a true half-part either (a) keep both halves and apply **symmetry BCs** on the XY cut face (ZSYMM if the face normal is Z), or (b) remove one half at the base feature via **Shape** → **Cut** → **Revolve**. Most analysts keep the full geometry + symmetry BC.

6. Isolate each hole for a structured mesh

1. Around each 7" hole, add a small **O-grid/"butterfly"** using **Partition Face** → **Sketch** (a square or octagon offset ring around the hole) so the hole rim sits in 4–8 quad faces.
 2. Keep every partition cell **simple (≤ 6 faces)** so it will accept Hex.
 3. The §3 rings + §4 spokes already frame each hole; the O-grid finishes the local pattern.
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7. Mesh controls

1. **Mesh** → **Assign Mesh Controls**: well-shaped cells → **Hex**, **Structured/Sweep**; leave transition cells **Tet** if needed.
 2. **Element type**: **C3D8I** or **C3D20R**, with ≥ 2 –3 elements through the 3.5" wall.
 3. **Seed hole edges** (≥ 8 –12 elements around each 7" hole), bias finer toward rims.
 4. **Mesh** → **Verify** (Analysis checks): watch aspect ratio, min angle, Jacobian near holes.
 5. Apply the correct **symmetry BC** on the XY cut face.
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8. End-to-end command summary

#	Action	Menu path
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1	Datum axis (centerline)	Tools → Datum → Axis → Principal axis (Y)
2	Datum plane (meridian/symmetry)	Tools → Datum → Plane → Principal plane (XY)
3	Section sketch (trace dome outline + hole-center line)	Tools → Partition → Cell → Sketch on XY plane
4	Shell revolve the section	Tools → Partition → Cell → Extrude/Sweep edges , sweep along circular edge (360° full / 180° half)
4alt	Ring fallback	Tools → Datum → Plane → Offset (XZ) at hole heights → Partition Cell → Use datum plane
5	Radial sectors to each hole center	Tools → Partition → Face → Shortest path (apex → hole center)
5alt	Sectors through thickness	Tools → Datum → Plane → Rotate about axis (per hole angle) → Partition Cell → Use datum plane
6	Half cut (last)	Tools → Partition → Cell → Use datum plane (XY)
7	Hole O-grid	Partition Face → Sketch

9. Common pitfalls

- **Extrude/Sweep translated instead of revolving:** you picked the *axis as a direction*. Pick a **circular edge as the path** to revolve.
- **360° on a half part fails:** a half model has only 180° of material — sweep the **semicircular** edge, or do the revolve **before** the half cut.
- **Sketch on the curved face:** the section sketch must sit on the **flat XY meridian plane**, not the curved dome surface. Use

Project Edges to trace the dome outline into that sketch.

- **Hole-center line misses holes:** confirm the traced line sits at the true **mid-thickness ($\approx 1.75''$)** and at each bolt-circle radius R_i ; a line drilled off-center won't pass through the hole centers.
 - **Too many sector planes:** for staggered patterns, reuse rotated datum planes where hole angles repeat between rings; otherwise prefer the **face shortest-path** method to avoid a plane per hole.
 - **Non-hex-able cells:** a cell that won't take Structured/Sweep usually has >6 faces — add one simplifying partition or set it Tet.
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Sources

- [Methods for partitioning cells — Abaqus/CAE Reference \(MIT mirror\)](#)
- [Using the sketch planar partition method to partition a cell — Abaqus/CAE](#)
- [Geometry Partitioning in Abaqus/CAE — TECHNIA](#)
- [Abaqus-CAE: Radial Partition — Eng-Tips forum](#)